Valentine Golden Ghanem

Principal Medical Laboratory Scientist · Computational Epidemiologist · Public Health Researcher

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Research Profile

Principal Medical Laboratory Scientist and computational epidemiologist with more than ten years of clinical laboratory leadership in medical microbiology and postgraduate training in data science, public health and law. Combines ensemble machine learning (XGBoost, Random Forest) with spatial statistics (Local Indicators of Spatial Association, LISA) to model infectious disease incidence and map health disparities across Ghana.

Research Interests

Epidemiology	Spatial analysis of infectious diseases; health inequities; social determinants of health
Data Science	Machine learning and explainable AI (XGBoost, decision trees, SHAP) for disease forecasting
Global Health	Health systems strengthening; global health law; cost-effectiveness of mHealth interventions

Education

Expected 2026 **Master of Laws (LLM), International Law**
Liverpool John Moores University, United Kingdom
Dissertation: Pathogen Data Sovereignty and One Health Surveillance: Ghana's Legal Duties under Global Health Security Law

2026 **Master of Science (MSc), Public Health**, with Merit
University of Suffolk, United Kingdom
Specialisation: Epidemiology and Non-Communicable Disease (NCD) Control
Dissertation: Cost-Effectiveness of mHealth Care Models for Hypertension in Ghana

2025 **Master of Science (MSc), Data Science**, with Distinction
University of East London, United Kingdom
Specialisation: Machine Learning and Spatial Analytics
Dissertation: Ensemble ML Modeling for HIV/AIDS Incidence Forecasting

2013 **Bachelor of Science (BSc), Medical Laboratory Science**, Second Class (Upper Division)
University of Ghana, Accra, Ghana
Specialisation: Medical Microbiology

Professional Experience

2015 – Present **Principal Medical Laboratory Scientist (Supervisory) and Epidemiology Researcher**
Cocoa Clinic, Ghana COCOBOD Medical Department, Accra, Ghana

- Integrate laboratory-confirmed disease surveillance data into the District Health Information Management System (DHIMS2), Ghana's national reporting platform.
- Build machine learning pipelines (XGBoost, Random Forest) on sub-national data to quantify the socio-behavioural and environmental drivers of infection.
- Supervise diagnostic operations in high-throughput laboratory units processing more than 30,000 samples per month under an ISO 15189 quality management system.
- Lead mobile screening outreach providing preventive care and diagnostics to more than 4,000 underserved people across regional clusters.

2014 – 2015 **Field Diagnostic Coordinator (Volunteer)**
Life Medical Mission, Accra, Ghana

- Coordinated community health outreach, managing mobile laboratory infrastructure and running rapid screening for endemic conditions for more than 5,000 beneficiaries.

Publications

Peer-reviewed journal articles (most recent first)

[3] **Ghanem, V. G.** (2026). Spatial and machine learning analysis of district-level health insurance inequities in Ghana. *Cureus*, *18*(1), Article e101984. https://doi.org/10.7759/cureus.101984
Key finding: Across 261 districts, bivariate LISA found 42 high-high uninsurance clusters; a decision tree (82.5% accuracy) ranked illiteracy as the primary predictor (84.7% feature importance).

[2] **Ghanem, V. G.** (2025). Socio-behavioral and spatial determinants of HIV/AIDS incidence in Ghana: An ecological cross-sectional study with explainable machine learning. *Advances in Infectious Diseases*, *15*(4), 669–694. https://doi.org/10.4236/aid.2025.154050
Key finding: In a region-level ecological analysis, counterfactual simulation projected that a 10% increase in educational access or HIV awareness could reduce HIV incidence by up to 16%.

[1] **Ghanem, V. G.** (2025). HIV prognosis in sub-Saharan Africa: A narrative review of mortality trends, structural determinants, and policy implications. *International Journal of Medical and Health Research*, *3*(2), 95–110. https://doi.org/10.61424/ijmhr.v3i2.346
Key finding: Synthesised 218 articles; only 43% of studies included actionable policy provisions.

Honours, Awards and Service

2026 **Alumni Achievement Award** (Diversity and Industry of the Future categories)
University of East London, United Kingdom

2026 **Unicaf Africa Prize for Academic Excellence**
For outstanding academic performance and multidisciplinary research leadership during the MSc Public Health, University of Suffolk

Service **International Partnerships Representative**, Alumni Advisory Board
University of East London, United Kingdom

Skills

Methods Biostatistics; machine learning (Python, R); spatial analysis (QGIS, LISA); SQL

Tools and data Tableau; Streamlit; Plotly Dash; DHIMS2; LIMS (LabWare, Meditech); Ghana Statistical Service census data extraction; Git (GitHub, GitLab)

Laboratory Medical microbiology; infectious disease diagnostics; ISO 15189 quality management systems

Professional Registration and Memberships

Registration Registered Medical Scientist, CORU (Ireland), No. MS 063958
Licensed Medical Laboratory Scientist, Allied Health Professions Council, Ghana, No. 09007815
Registered Epidemiologist A and Member, Vereniging voor Epidemiologie (VvE), Netherlands

Fellowship Fellow, Royal Society for Public Health (FRSPH), United Kingdom, No. 140437

Membership Academy of Clinical Science and Laboratory Medicine (ACSLM), Ireland, No. 12474
Ghana Association of Biomedical Laboratory Scientists (GABMLS), No. AS 1206

In progress Certified in Public Health (CPH), National Board of Public Health Examiners, December 2026